RED SEAL EXAM PREP

420B Ironworker (Generalist)

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Grouped by this site's own topic areas
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Mock Exam — answer all questions, then check the key

Q1. **During a crane lift, what is the purpose of a tagline?**

A) To control the rotation and swing of the load

B) To measure the length of the lift path

C) To attach the load to the crane hook

D) To provide an extra rigging connection in case the main sling fails

Q2. **What is the yield strength of Grade 400W rebar per CSA G30.18?**

A) 500 MPa

B) 600 MPa

C) 300 MPa

D) 400 MPa

Q3. **Synthetic web slings must be removed from service when:**

A) There are cuts, tears, or chemical burns visible

B) The sling has been used for more than 500 lifts

C) Only when the rated capacity tag is illegible

D) Colour fading or minor surface abrasion is visible

Q4. **Two structural steel beams are to be connected to a common column. Under Ontario's Construction Projects regulation, how may that connection be made?**

A) With both beams sharing one set of bolts through the column

B) As a clipped double connection, or the first beam seated

C) With the second beam tack welded until the bolting crew comes

D) With the first beam's bolts pulled to let the second one in

Q5. **A "column splice" in a steel structure is located:**

A) At the top of the column where the roof beam connects

B) At an intermediate height where two column pieces meet

C) At the beam-to-column connection at each floor level

D) At the foundation level where the column meets the base plate

Q6. **Under AWS D1.1, when welding ASTM A572 Grade 50 steel with low-hydrogen SMAW electrodes, the minimum prequalified preheat and interpass temperature for material over 38 mm up to 65 mm thick is approximately:**

A) 107°C (225°F)

B) No preheat required

C) 66°C (150°F)

D) 10°C (50°F)

Q7. **A column dowel extends from the footing to lap splice with the column bars. The minimum lap splice length is typically:**

A) Determined by the engineer's calculations

B) Always 300 mm for all bar sizes

C) Equal to the concrete cover thickness × 10

D) Equal to the bar diameter × 10

Q8. **A beam has been landed in its connection and the connectors are ready to cut it loose from the crane. What has to be in that connection at that moment?**

A) Two bolts run on by hand so the beam can still be shifted

B) Enough bolts to stop rotation, drawn up wrench-tight

C) One bolt wrench-tight and a drift pin in a second hole

D) Every bolt run in and torqued to the design drawings

Q9. **On a structural drawing, what does the note "FSBW" refer to?**

A) Field Shop Bolt Work

B) Full-Strength Butt Weld

C) Fillet Strength Below Weld line

D) Fabricated Steel Bearing Wall

Q10. **No manufacturer's specification can be found for a winding drum that a wire rope will spool onto. What minimum drum diameter should be used?**

A) 20 times the rope diameter

B) 6 times the rope diameter

C) 30 times the rope diameter

D) 12 times the rope diameter

Q11. **What does the abbreviation "TOS" mean on a structural steel drawing?**

A) Top Of Steel — the elevation of the beam top flange

B) Total Overall Span — the beam span between columns

C) Tension On Steel — a notation for tension member design

D) Type Of Section — indicating the steel section type

Q12. **A "burning bar" (thermic lance) is used in ironwork for:**

A) Heating structural steel for straightening or hot bending

B) Marking cut lines on structural steel by burning shallow grooves

C) Preheating weld areas to prevent hydrogen cracking

D) Cutting hard materials like steel, concrete, and rock

Q13. **An ironworker is welding a moment connection on a seismic force-resisting frame. What additional requirement applies compared to a regular gravity connection?**

A) All welds must be made using GTAW process

B) The connection must be welded vertically-up only

C) Higher amperage settings must be used to achieve deeper penetration

D) Notch-tough weld metal with CVN requirements

Q14. **An ironworker is about to work on fall arrest at the top of a column. What must the employer have in place before that system is used at all?**

A) Written rescue procedures for a worker whose fall is arrested

B) A second worker at the base ready to lower the crane's hook

C) A signed waiver acknowledging the risk of suspension trauma

D) A spare harness and lanyard staged at grade for the rescuer

Q15. **On a structural drawing, the connection dimensions for a diagonal brace are run from a mark labelled W.P., the working point. Unless the detail states otherwise, what does that mark locate on the joint?**

A) The corner of the gusset plate nearest the column face

B) The point where the brace edge crosses the beam top flange

C) Where the centrelines of the connected members intersect

D) The centre of the bolt group joining the brace to the gusset

Q16. **Under the Alberta OHS Code, what is the maximum free-fall distance permitted for a personal fall arrest system that has no shock absorber?**

A) 0.6 m (2 ft)

B) 3 m (10 ft)

C) 1.2 m (4 ft)

D) 1.8 m (6 ft)

Q17. **Standard rebar ties are made using:**

A) Epoxy adhesive at each bar crossing

B) Thin plastic zip ties rated for 50 kg

C) 16-16.5 gauge soft annealed steel wire

D) Spot welding at each intersection

Q18. **An ironworker is welding a moment connection on a structural beam using E71T-8 flux-cored wire. The WPS specifies a minimum preheat of 150°C. The ambient temperature is -5°C and the base metal temperature is 4°C. What must be done before welding?**

A) Begin welding immediately — preheat is for thick plate only

B) Increase wire speed to compensate for cold metal temperature

C) Preheat the joint to 150°C and verify with temp sticks

D) No preheat needed when ambient is above freezing (-5°C is the ambient air, not metal)

Q19. **What weld symbol indicates a fillet weld?**

A) A right triangle on the reference line

B) A wavy line below the reference line

C) A square groove symbol

D) A circle on the reference line

Q20. **A crane is lifting a 20,000 kg load at a 15m radius. The load chart shows maximum lift capacity at 15m is 18,000 kg (with all outriggers fully extended on firm ground). What should the ironworker signalperson communicate to the operator?**

A) Proceed if the crane manufacturer approves in writing

B) Proceed if the lift will be completed in under 2 minutes

C) Proceed — crane charts include a 10% safety margin for occasional overloads

D) Stop the lift — it exceeds the chart capacity

Q21. **What is a "beam clamp" used for in ironworking?**

A) To permanently connect two beams at a splice

B) To attach rigging to a beam flange without welding

C) To clamp the welding electrode holder to the beam during welding

D) To hold a beam in plumb during erection until it is bolted

Q22. **On an Ontario construction project, a mobile crane is lifting near an energized overhead distribution line with a nominal phase-to-phase voltage of 44 kV. The line owner has not installed protective devices or set up its own procedures. Under Ontario's Construction Projects regulation, what is the minimum distance that any part of the crane, its rigging or its load may be brought to the line?**

A) 4.5 m

B) 1.5 m

C) 6 m

D) 3 m

Q23. **What is the purpose of a base plate in a steel column installation?**

A) To prevent corrosion of the column base

B) To hold the anchor bolts

C) To allow lateral adjustment after erection

D) To spread the column load over the foundation

Q24. **Open-web steel joists have been landed and both bearing ends attached, but the bridging is only partly installed. What may be placed on them?**

A) Deck bundles, provided each one is landed over a bearing end

B) Loads up to half of the joists' design capacity may be placed

C) Anything the joists were designed to carry, once both ends bear

D) No construction load at all until the bridging is fully anchored

Q25. **A rigid steel weldment is picked on a four-leg bridle sling. Under British Columbia's OHS Regulation, the working load limit of the whole sling is limited to the working load limit of how many of its legs?**

A) Two of them, the pair on the same diagonal

B) Whatever number a professional engineer sets

C) All four, since the four legs share the load

D) Three of them, whichever three are chosen

Q26. **When installing high-strength bolts using the turn-of-nut method, what is the first step?**

A) Install all bolts finger-tight only

B) Apply lubricant to all bolt threads

C) Bring all bolts to snug-tight first

D) Apply full final torque immediately

Q27. **What do the deformations (ribs) on deformed rebar do?**

A) Reduce the bar cross-section to save material at low-stress locations

B) Mechanically interlock with concrete to improve bond and force transfer

C) Increase the bar weight to improve resistance to impact loads

D) Provide identification markings for the steel mill

Q28. **A weld inspection reveals "undercut" along the toe of a fillet weld. What caused this?**

A) Excessive amperage or travel speed

B) Insufficient preheat temperature

C) Electrode too small for the fillet size

D) Moisture in the electrode coating

Q29. **An anchor rod at a column base is 25 mm out of position. Which statement about correcting it in the field is correct?**

A) Reaming the base plate hole oversize keeps the rod's capacity

B) Cutting it off and welding on an extension is a standard field fix

C) Bending it cold with a rod bender restores its design strength

D) Heating or flame-cutting the rod changes its strength properties

Q30. **During a lift, a worker who is not the designated signaller sees a hazard and gives the crane operator a STOP signal. Under the occupational health and safety rules in Alberta and British Columbia, what must the operator do?**

A) Stop the operation as soon as the signal is seen

B) Land the load first, then act on the STOP signal

C) Stop only if the designated signaller repeats it

D) Carry on; only the designated signaller directs it

Q31. **What is a "lap splice" in reinforcing steel?**

A) A welded joint fusing two rebar pieces at their ends

B) A mechanical coupler that joins two rebar end-to-end

C) An overlap where two parallel bars transfer stress

D) A bar bent into a U-shape hooked around another bar

Q32. **An ironworker is placing shear studs on a composite steel beam with a drawn-arc stud gun. Studs from the run are bend tested as a check on weld quality. One stud is bent about 30 degrees off its axis and a crack opens in the weld at the base of the stud. What is the required action?**

A) Accept it - minor bend-test cracks are normal

B) Replace the stud and raise the test rate

C) Cut out and re-weld only the cracked stud base

D) Reject the beam and replace every stud on it

Q33. **A horizontal lifeline is rigged between two anchors 10 m apart and sags 0.3 m at midspan. A 100 kg worker falls from midspan. What can be said about the force at the anchors?**

A) Anchor force is the worker weight plus the lanyard weight, and no more

B) Each anchor carries exactly 490 N, which is half the worker body weight

C) The lifeline cancels anchor force because the load hangs straight down

D) The anchor force is many times the worker weight because sag is shallow

Q34. **An ironworker welds out a multi-pass fillet weld on a structural connection in cold, damp weather. The base metal is A36 and the filler is E7018. The weld passes visual inspection at the end of the shift, but the next morning cracks are found at the weld toe and in the heat-affected zone. What is the MOST likely cause?**

A) Arc voltage is too high, causing undercutting at the toes

B) Hydrogen cracking from moisture in the electrode

C) Travel speed is too slow - too much heat input

D) Base metal carbon content is too high for E7018

Q35. **An ironworker 8 m above grade must temporarily remove a section of guardrail to swing a beam into position, and no overhead tie-off point exists at that spot. What must happen before the guardrail comes off?**

A) Nothing more, because the opening will be left open for five minutes

B) An entry in the site log book recording that the guardrail came off

C) Alternate protection at least equal to the guardrail is in place

D) A signaller below keeping everyone out of the area beneath the work

Q36. **A wire rope sling is rated for a working load limit (WLL) of 10,000 kg in a straight-line (vertical) hitch, and the sling tag shows no separate choker rating. What is the effective WLL in a choker hitch at an angle of choke of 120° or greater?**

A) 5,000 kg — choker hitch reduces capacity by 50%

B) 10,000 kg — choker hitch does not reduce capacity

C) 8,000 kg — choker hitch reduces capacity by 20%

D) 7,500 kg — choker hitch reduces capacity by 25%

Q37. **ASTM F3125 Grade A325 and Grade A490 are the two common high-strength structural bolt grades. Which statement about them is correct?**

A) A490 may be reused after full pretensioning; A325 may not

B) Both require the same minimum pretension at a given size

C) A325 is stronger than A490 and used for most connections

D) A490 requires a higher minimum pretension than A325 does

Q38. **An ironworker is rigging a steel plate that has sharp flame-cut edges, and a polyester web sling will bear directly on one of those edges. What has to be done before the plate is lifted?**

A) Double the sling so the added ply takes the cutting action

B) Fit corner protectors between the plate edges and the sling

C) Take up the load slowly so the edge cannot saw the webbing

D) Cut the rated load in half and lift the plate as it is rigged

Q39. **Rebar arrives on site carrying tight mill scale and light surface rust. What should the ironworker do before it is placed?**

A) Apply a rust inhibitor coating before placing in forms

B) Place the bars as is — light rust is acceptable

C) Replace all lightly rusted bars with new material

D) The bars must be sandblasted clean before placement

Q40. **On a structural drawing, what does the symbol "EL. 12500" typically mean?**

A) The point's elevation above the project datum

B) Element number 12,500 on the fabrication list

C) The element has a design load of 12,500 kg

D) The element is 12,500 mm in overall length

Q41. **What does a "Personal Fall Arrest System" (PFAS) consist of?**

A) Safety net and horizontal lifeline only

B) Safety boots, hard hat, and safety glasses

C) Body belt, rope grab, and vertical lifeline

D) Full body harness, connecting lanyard, and anchor point

Q42. **What is the difference between a "drift pin" and a "spud wrench"?**

A) They are the same tool — different regional names

B) A drift pin can tighten bolts; a spud wrench cannot

C) A spud wrench is used only for rigging; a drift pin is used only for concrete work

D) A spud wrench adds a wrench end; a drift pin only aligns

Q43. **An ironworker inspects a 6×19 IWRC wire rope sling before a critical lift and finds 11 randomly distributed broken wires in one rope lay. What action should be taken?**

A) Remove the sling from service immediately

B) The sling may be used if the broken wires are not load-bearing

C) Tag the sling for monitoring and continued use

D) The sling may be used — the limit is 20 broken wires per rope lay

Q44. **An ironworker is placing rebar in a concrete footing. The drawing calls for 20M bars at 200mm c/c each way, top and bottom, with 75mm cover to every face, and shows the footing 200mm thick. After placing the bottom mat, the ironworker finds the top mat cannot fit within the specified cover. What is the CORRECT response?**

A) Reduce the bottom cover to 50mm to create space for the top mat

B) Place both mats touching - concrete cover can be eliminated for interior footings

C) Eliminate the top mat - one layer of rebar is sufficient for footings

D) Notify the superintendent and engineer of the conflict

Q45. **A structural drawing shows "4-M20 A325 bolts" at a beam-to-column shear tab connection. What does this mean?**

A) A bolt group with a shear capacity of 4,000 N per M20 bolt

B) Metric bolts, Group 4 size class, rated 20 kN shear capacity each

C) 4 machine bolts of 20 mm diameter, Grade A325 galvanized finish

D) 4 bolts, metric M20 (20 mm diameter), ASTM A325 high-strength

Q46. **What does CSA W47.1 govern?**

A) The testing and inspection of completed structural welds

B) The certification of fusion welding companies in Canada

C) The classification of welding consumables

D) The design of structural steel connections

Q47. **What does "100% tie-off" mean for ironworkers during structural steel erection?**

A) The ironworker must be tied off for 100% of the workday

B) 100% of the rigging must be secured before erection begins

C) The ironworker must use a full 100-foot lifeline at all times

D) At least one PFAS connection at all times, even when moving

Q48. **Before using any rigging hardware (shackles, hooks, chains) on a job site, the ironworker should:**

A) The equipment only requires inspection during the annual company audit

B) Visually inspect for damage and verify adequate WLL

C) Test each piece by applying twice the WLL to check for deformation

D) Apply a fresh coat of oil to all moving parts

Q49. **A beam detailed with 25 mm of camber arrives on site with its camber paint mark worn off. How does the crew establish which way up it goes before it is flown?**

A) Sight along the beam on the ground to find its natural bow

B) Measure the depth at both ends and put the deeper end up

C) Fly it either way, since 25 mm is inside erection tolerance

D) Assume the mill mark stamped on the web is meant to face up

Q50. **A column is being lowered onto its anchor rods. What is the correct practice for the connectors receiving it?**

A) Steady the base plate by hand so the holes drop over the rods

B) Guide it with taglines, hands and feet clear of the base plate

C) Foot the base plate from the footing to stop it from swinging

D) Slack the load line so the column settles onto the rods freely

Answer Key & Explanations

> Scoring: 35 correct out of 50 is 70%. These 50 questions come from the 121-question 420B bank; the last page shows what the complete printable bank contains.

Q1. Answer: A

Taglines allow workers to guide and control a suspended load from a safe distance, preventing rotation and swing without placing workers directly under or near the load.

Q2. Answer: D

Grade 400W rebar has a minimum specified yield strength of 400 MPa and is weldable (the "W" designation). It is the most common structural rebar grade in Canadian construction.

Q3. Answer: A

A synthetic web sling comes out of service for damage that has cut, burned, melted or abraded the fibres or the stitching. Alberta's OHS Code s.305(2) requires permanent removal if part of the sling is melted, charred or damaged by chemicals, if stitches in load-bearing splices are broken or worn, or if end fittings are excessively pitted or corroded, cracked, distorted or broken. Section 305(1) adds the measured limits: an edge cut longer than the web thickness, or abrasion deeper than 15 percent of the webbing thickness taken across all plies. Colour fading and light surface scuffing are not by themselves removal criteria. No regulation retires a sling on a lift count. An illegible or missing capacity tag does require removal, but it is one criterion among many rather than the only one.

Q4. Answer: B

Ontario Regulation 213/91, s. 31 (2) allows two arrangements and no others: if two structural steel columns or structural steel beams are connected to a common column or common beam, the connection shall be made using a clipped double connection, or the first column or beam shall be secured in a seated connection. Both arrangements do the same job - the first member keeps support of its own, so hanging the second member does not depend on disturbing what is holding the first. That is what makes the other answers the wrong instinct. Pulling the first beam's bolts to make room takes the support out from under a member that a connector may be standing on. A tack weld is a fit-up aid rather than the connection the drawings call for, and it is not one of the two arrangements the section permits. Running both beams on one common set of bolts leaves the two members depending on the same fasteners, which is the arrangement the section is written to prevent.

Q5. Answer: B

A column splice is the joint between an upper column section and the one below it, at a height where a single piece would be impractical to fabricate, ship or handle. It is set above a floor level so the connection can be reached and worked from the floor, clear of the beam-to-column connections, and away from points of maximum stress; the splice elevations for any given job are shown on the erection drawings. That is what separates it from the other three answers, which are all real locations on a column but none of them a splice: the beam-to-column connections at each floor are framing connections, the top of the column is where the roof framing lands, and the foot of the column is the base plate on the anchor rods. Ontario's O. Reg. 213/91 uses the splice as a marker for working height. Section 33 (1), subject to s. 33 (2), says work on a building shall not be carried out higher than the higher of two storeys or the first column splice above the temporary or permanent flooring, and s. 33 (2) allows three storeys where the vertical distance between the tiers of column splices exceeds two storeys. Section 33 (3) takes the section off a worker who is on a scaffold, a worker whose fall would be arrested by a safety net without endangering the worker, and a worker using a fall arrest system attached to the project, so on a job where the crew is tied off to the structure this is not the limit that governs.

Q6. Answer: C

AWS D1.1 places ASTM A572 Grade 50 in Category B, and Category B welded with low-hydrogen SMAW electrodes calls for 150°F (66°C) on material over 38 mm through 65 mm thick. The full Category B row of the prequalified preheat table (Table 5.8 in the 2020 and later editions; Table 3.2 in earlier ones) reads: 3 to 20 mm = 32°F (0°C); over 20 through 38 mm = 50°F (10°C); over 38 through 65 mm = 150°F (66°C — the metric column of D1.1M rounds this to 65°C); over 65 mm = 225°F (107°C). So 107°C is correct only for plate thicker than 65 mm, and that is the trap in this question. Watch the band edges too: a section exactly 38 mm thick still falls in the 10°C band, because the row reads 'over 38'. On Canadian structural work AWS D1.1 is not the governing document. Building Code Article 4.3.4.1 requires buildings and structural members made of structural steel to conform to CAN/CSA S16, and S16 in turn requires a CSA W47.1 certified fabricator with joint design and quality to CSA W59 — so preheat comes off the CWB-accepted WPS/WPDS, never off a US table from memory. The Canadian counterpart grade is CSA G40.21 350W (350 MPa minimum yield, against 345 MPa for A572 Grade 50), which is what a Canadian mill certificate will normally show.

Q7. Answer: A

Dowel lap splice length is calculated by the engineer based on development length calculations considering bar size, grade, concrete strength, cover, and splice class. Ironworkers must follow the drawings — never assume a standard splice length.

Q8. Answer: B

IHSA's ironworker chapter puts it in one line: before being cut loose, the beam must be bolted so that it will not rotate, at least two or more bolts should be put in position depending on the engineer's requirements, and a drift pin or a wrench placed in a hole is not to be relied on. Two things therefore have to be true at that moment, and the count is only one of them. Enough fasteners have to be in to take rotation out of the connection, and IHSA's structural steel erection guide supplies the condition that goes with them: bolted and temporary connections are to be wrench-tight, and are torqued later, to the design specification, during bolt-up and detail work. Bolts run on by hand leave the beam free to move in the connection, which is the state the rule exists to prevent. A drift pin is a tool for drawing holes into line; it carries no load and can walk out, which is why the guidance names it as the thing not to trust. Waiting for the whole connection to be torqued keeps the crane tied to one beam - the bolting-up crew follows the erection crew and installs the remaining bolts afterward. How many bolts is not the connector's own call: the number comes from the engineer's requirements and the erection plan, and the same guide sets one bolt per connection for diagonal bracing unless the structural design engineer specifies otherwise, which is a different member and a different number.

Q9. Answer: B

FSBW (Full Strength Butt Weld) or CJP (Complete Joint Penetration) groove weld indicates that the weld must develop the full tensile strength of the connected base metal — the weld is not the weak link. This requires 100% penetration through the joint thickness, usually verified by visual inspection plus UT (ultrasonic testing) or radiography. It is different from: a partial joint penetration (PJP) weld, which does not penetrate the full thickness and has reduced capacity; a fillet weld, which joins members at a surface (no groove); and a plug or slot weld. FSBW/CJP welds are required at moment connections (beam-column moment joints) and splice plates carrying tension — they cost more (more prep, more passes, more inspection) than fillet welds.

Q10. Answer: A

The binding rule is the manufacturer's figure: Alberta's OHS Code s.302(1) requires that the wire ropes, sheaves, spools and drums used in rigging have a diameter of not less than the diameter specified by the manufacturer for that use. Where no specification can be found, Alberta's published explanation guide to Part 21 gives the working minimum — unless otherwise specified by the manufacturer, the diameter of winding drums should not be less than 20 times the diameter of the rope, which reduces the likelihood of weakened rope caused by excessive bending stresses. Six or twelve times the rope diameter bends the rope far harder than it is built to take; thirty times is generous but is not the published minimum. Bending fatigue, not overload, is what usually kills a rope. Note that there is no single number for a rope running over a sheave: the minimum varies with rope construction, so read the rope manufacturer's table rather than carrying one figure in your head.

Q11. Answer: A

TOS (Top of Steel) is an elevation reference on structural drawings indicating the height of the top flange of a beam, measured from the project datum. It is used to coordinate with the concrete deck, mechanical, and architectural drawings. Common related notations: BOS (Bottom of Steel — soffit elevation, important for clearance), TOS is NOT the same as finished floor elevation (which is TOS + deck slab thickness + topping if any). Nor is TOS the beam depth or the centreline elevation. For steel erection: when setting beam elevation, you level to the TOS — not to the bottom flange or web centreline.

Q12. Answer: D

A burning bar, or thermic lance, is a steel tube packed with steel or alloy wires and fed with oxygen; once lit, the iron itself burns in the oxygen stream and the bar washes its way through the work. The reaction is hot enough to melt concrete and rock as well as steel, so a burning bar gets through material an oxy-fuel torch cannot cut at all - heavily reinforced concrete, very thick or hardened sections, seized pins and frozen anchor bolts - which is why it turns up in demolition and emergency access work. It is not a preheating or heat-straightening tool: those call for a rosebud or heating torch with controlled, even heat, and a burning bar destroys the material instead of warming it. It does not lay out cut lines either. The bar is consumed as it burns and throws molten slag, so it demands hot work precautions and a fire watch.

Q13. Answer: D

Seismic connections take large cyclic deformations and have to absorb that energy by yielding rather than by fracturing. The weld metal must therefore be notch tough, which is proven by Charpy V-notch (CVN) impact testing at the low temperature and minimum absorbed energy called up by the project specification and the accepted welding procedure. For the same reason the procedure normally sets a maximum heat input and a maximum interpass temperature, because too much heat coarsens the grain and costs the weld its toughness. In Canada the design of seismic force-resisting systems is governed by CSA S16, referenced through the National Building Code; the American equivalents, AISC 341 and AISC 358, apply only on a project whose specification calls them up.

Q14. Answer: A

Ontario Regulation 213/91 s. 26.1 (4): "Before any use of a fall arrest system or a safety net by a worker at a project, the worker's employer shall develop written procedures for rescuing the worker after his or her fall has been arrested." The requirement is a precondition, not an afterthought - it bites before any use of the system, so a crew that ties off without a rescue plan is offside from the first connection. The reason is that arresting the fall is only half the event. A worker hanging in a harness is held by leg straps that restrict blood return, and suspension becomes dangerous in a time frame measured in minutes, so a rescue that has to be invented on the spot is a rescue that arrives late. Staging spare equipment, or posting someone at the base, may well form part of a procedure, but neither is the requirement and neither is any use without a plan that says who does what, with what, and how fast. A waiver is not a control and does not discharge a duty under the regulation.

Q15. Answer: C

The working point is where the centrelines - the gravity axes - of the members meeting at that joint intersect. The detailer sets the joint out from that point so that the lines of action of the column, the beam and the diagonal brace all meet in one place. When they do, the joint delivers axial force and no member picks up a bending moment from an offset. Shift the brace's line of action away from the working point and an eccentricity is introduced, and the moment that eccentricity produces has to be carried by the connection and by the members framing into it. The choice of working point belongs to the engineer, so a detail that deliberately works to a point away from the member centrelines says so on the drawing and the connection is designed for that eccentricity - which is why the stem says "unless the detail states otherwise".

For the crew the working point is a layout reference, not a piece of hardware. On a brace-to-beam-to-column joint it commonly falls where the column and beam centrelines cross, inside the members themselves, where there is nothing to fasten to; the gusset plate exists precisely because the members cannot all physically meet at that point. That is what separates it from the three wrong answers, each of which is a real feature you can put a tape on. The centre of the bolt group is where the brace is fastened to the gusset, and its position is set out from the working point rather than the other way round. The corner of the gusset and the place where the edge of the brace crosses the beam flange are edges of fabricated plate and shape; they move if the gusset is resized or the brace section is changed, while the working point stays where the member centrelines cross.

Q16. Answer: C

The Alberta OHS Code requires a personal fall arrest system without a shock absorber to limit a worker's free-fall distance to 1.2 m. A shock absorber is normally required in the first place; a system without one is allowed only where it meets that free-fall limit and the rest of the same section. Where a shock absorber is used, the permitted free fall is the maximum free-fall distance the absorber is labelled and rated for, and the system must still be arranged so the worker cannot strike the ground or a level below, and must limit the maximum arresting force on the worker to 6 kN, or 8 kN with an E6 shock absorber used to the manufacturer's specifications. The practical control on free fall is anchor height: the Code requires the lanyard to be secured to an anchor no lower than the worker's shoulder height, or as high as reasonably practicable.

Q17. Answer: C

Rebar ties are made from 16 to 16.5 gauge (approximately 1.6 mm diameter) smooth, soft annealed steel wire, twisted with a tier tool. Multiple tie patterns exist for different conditions (snap tie, saddle tie, figure-8 tie).

Q18. Answer: C

Preheat requirements are based on base metal temperature, not ambient air temperature. At 4°C base metal temperature with a 150°C preheat requirement, the joint must be heated to 150°C minimum with a heating torch before striking the arc. Cold steel increases the risk of hydrogen-induced cracking (HIC). Verify with temperature sticks or a contact pyrometer immediately before welding and maintain interpass temperature throughout.

Q19. Answer: A

A fillet weld is shown as a right triangle on the weld symbol, with the perpendicular leg always on the left and the hypotenuse facing outward (away from the joint). The weld size is shown to the left of the symbol.

Q20. Answer: D

Load chart values are maximums — a working lift must never exceed them. The margin between the chart rating and the crane's tipping or structural limit is not usable capacity. In Ontario, for example, rated capacity is determined to CSA Z150, and a crane may not be loaded beyond it except during load tests required by the manufacturer. Exceeding the chart rating creates an unacceptable risk of crane overturning. The lift must be stopped and re-planned: decrease the lift radius (boom in), reduce the load, or bring in a larger crane.

Q21. Answer: B

Beam clamps attach to a beam flange by clamping action, providing a temporary rigging point for hooks, slings, and blocks — for hoisting, pulling, or hanging equipment without welding, drilling, or otherwise permanently modifying the beam.

Q22. Answer: D

O. Reg. 213/91, s. 188(2) says no object may be brought closer to an energized overhead conductor than the distance set in its Table, based on the conductor's nominal phase-to-phase voltage. The first band covers 750 volts or more, up to 150,000 volts, and requires 3 m. A 44 kV line (44,000 V) falls in that band. The 4.5 m distance is only for lines over 150,000 V up to 250,000 V, and 6 m is only for lines over 250,000 V. Picking either one for a distribution line means using the wrong voltage band. The Table sets no distance smaller than 3 m for any line of 750 V or more, so 1.5 m would put the load inside the legal limit. When a crane or its load could come within this distance, s. 188(4) to (8) also apply. The constructor needs written measures and procedures, warning devices and a sign at the operator's station, and a competent worker assigned as signaller who can see both the conductor and the equipment. The distances vary by province, so this answer is for Ontario only.

Q23. Answer: D

A base plate distributes the concentrated column load over a larger bearing area on the concrete foundation, reducing bearing stress to within allowable limits.

Q24. Answer: D

IHSA's structural steel chapter states the rule without qualification: no construction loads on steel joists until all bridging is installed and anchored and all joist-bearing ends are attached. A bare open-web joist is a slender member with almost no lateral stiffness. Its design capacity is a figure that assumes the bridging is there to hold the top chord in line; until it is, the joist can roll or buckle sideways under a fraction of that number. That is why half the design capacity is the wrong instinct - the design capacity is not a valid reference point yet, so no fraction of it is safe either. The same reasoning rules out the deck bundle answer: bundles are exactly the construction load the rule is about, and the same chapter requires them to be landed on fully bolted framing beside a main supporting beam at a column or wall, not on joists that are not yet braced.

Q25. Answer: D

WorkSafeBC's OHS Regulation s. 15.33 (2) is one sentence: the working load limit of a sling with more than three legs is limited to the working load limit of any three legs of the sling. A fourth leg is allowed on the hook; it simply cannot be counted. The mechanics behind that are worth carrying to any jurisdiction. Four legs holding one rigid piece are statically indeterminate - there are more supports than the statics of the lift can define - so which legs are actually tight is decided by millimetres of leg length and by exactly where the pick points sit, not by symmetry. Nothing about the arrangement guarantees that all four come up tight together, so adding four leg capacities describes a lift that may not happen. The section does not stop at two, though: three is the cap it states. The subsection names no engineer: it sets the cap itself and does not hand the number to anyone to reset. Two further limits ride along in the same section: the working load limit of no individual component of the assembly may be exceeded (15.33 (1)), and the load carried by any single leg of a bridle must not exceed the working load limit of that leg (15.33 (3)).

Q26. Answer: C

Turn-of-nut requires all bolts in the connection to be brought to the snug-tight condition first (full effort of an ironworker with a spud wrench), then a specified rotation is applied to achieve the minimum pretension.

Q27. Answer: B

The transverse and longitudinal ribs on deformed bar create mechanical interlock with the surrounding concrete, dramatically improving bond strength compared to smooth bars.

Q28. Answer: A

Undercut is a groove melted into the base metal at the weld toe that is not filled by weld metal. It comes from too much heat at the toe — excessive amperage, too fast a travel speed, too long an arc, or an electrode angle that throws the arc onto the vertical member instead of into the corner. It matters because it removes section from the base metal and leaves a sharp notch at the toe, which is where fatigue cracks start. The other faults listed cause different problems: moisture in the coating causes porosity, too little preheat on heavy or restrained sections invites cracking, and too small an electrode gives an undersized weld rather than a gouged toe.

Q29. Answer: D

Heat is the thing to be careful of. An anchor rod's strength comes from its steel chemistry and, in the higher grades, from heat treatment; taking a torch to it drives the material through temperatures that alter those properties, and flame-cutting leaves a heat-affected zone and a reduced section in a member designed to carry tension. Neither effect is visible afterward, which is what makes it dangerous - the rod looks fine and is no longer the rod the engineer designed. That is why heating and cutting are not the erector's call, and why any remedy for a misplaced rod comes back with engineering direction. The other three are field fixes, and none of them is sound. Cold bending does not restore design strength; it works the steel, leaves residual stress at the bend and puts a permanent kink where the load path wanted a straight bar. Welding an extension onto a cut rod assumes the rod steel is weldable and that a field splice can carry the design tension, and neither can be assumed. Reaming the plate hole oversize changes how load transfers between rod and plate and can leave the rod bearing on very little, which is why oversized holes at base plates carry their own plate washer requirements.

Q30. Answer: A

A STOP signal is the one signal an operator takes from anyone. Alberta's OHS Code has the operator take signals only from the designated signaller where one is designated, and then makes the exception in section 191(7): despite that, an equipment operator must take a STOP signal from a worker who is not a designated signaller. British Columbia's OHS Regulation, section 14.47(2), says the same for cranes and hoists: the operator must stop the operation of the equipment on receiving a stop signal from any person, and CCOHS puts it as obeying a STOP signal no matter who gives it. Waiting for the designated signaller to repeat the signal, carrying on because only that signaller directs the crane, or landing the load before acting on the STOP all apply the designated-signaller rule to the one signal both regulations take out of it. The hand signals themselves are not universal: published charts differ (CCOHS shows STOP as the arm extended, palm down, held rigidly, and EMERGENCY STOP as the same arm with the hand moved rapidly right and left), and Ontario's construction regulation requires a telecommunication system or prearranged visual signals, so use the signal set agreed for the site.

Q31. Answer: C

A lap splice is an overlapping zone where two bars run parallel: the bar ends overlap by a specified length (development length × splice factor) so that bar forces transfer through bond with the surrounding concrete.

Q32. Answer: B

A crack in the weld during a bend test is a failed test, not a normal result. Bending the stud off its axis loads one side of the weld collar in tension, and a sound drawn-arc weld takes that without cracking. A crack means the weld metal or the fusion at the base is unsound - commonly from moisture, paint, mill scale or oil on the flange, a damaged or damp ferrule, or wrong current or plunge time. The response is to replace the failed stud, removing it and making good the base metal before welding a replacement with corrected settings, and to widen the sample so the studs put down alongside it under the same conditions are proved as well. Dressing and re-welding the base of a suspect stud proves nothing about that stud, and one failure does not condemn every stud on the beam. How many studs are tested, and how far the sampling is widened after a failure, come from the project specification and the welding inspection procedure for the job.

Q33. Answer: D

A shallow sag turns a small vertical load into a very large line tension. Treat the line as two straight legs meeting at midspan: each leg rises only 0.3 m over the 5 m half span, so the legs are nearly horizontal and most of the tension acts along the line rather than upward. Static equilibrium at midspan gives tension ≈ (W × L) / (4 × sag), where L is the FULL span between the anchors, not the half span: (980 N × 10 m) / (4 × 0.3 m) ≈ 8,200 N, more than eight times the worker weight — and that is the static figure, before any dynamic effect of arresting a fall. Letting the line sag more lowers the tension; pulling it tight raises it sharply. That is why a horizontal lifeline and its end anchors are an engineered system and are never improvised on site.

Q34. Answer: B

Cracking that turns up after the joint has cooled, at the weld toe and in the heat-affected zone, is the signature of hydrogen-induced cracking - which is why the trade also calls it delayed or cold cracking. E7018 is a low-hydrogen electrode, but its coating draws moisture out of the air as soon as the hermetic package is opened. In the arc that moisture breaks down and puts hydrogen into the weld; the hydrogen then diffuses into the hard, highly stressed heat-affected zone and cracks it hours after the arc is out, once the joint has cooled to roughly 100 to 200 degrees Celsius. Cold, damp weather works against you on both counts - more moisture available, and a faster cooling rate that leaves a harder heat-affected zone. Keep opened low-hydrogen electrodes in a rod oven at about 120 degrees Celsius, recondition them once only, throw out any electrode that has been wet, and preheat the joint. Watch the timing, because it separates two different failures: a crack running down the centre of the bead during the pass or as you finish it is solidification, or hot, cracking. High arc voltage produces undercut, not cracking; A36 is a low-carbon steel well inside the range E7018 is made for; and slow travel with high heat input slows the cooling rate, which reduces hydrogen cracking rather than causing it.

Q35. Answer: C

The guardrail IS the protection at that edge, so taking it out creates an unprotected edge unless something equal to or better than the guardrail is already protecting the worker. The substitute has to be working before the rail comes off, not arranged afterwards: a personal fall arrest system connected to a temporary engineered anchor installed for the purpose, a safety net rigged below the work level, or a personnel platform that carries its own guardrails. If none of those can be provided at that location, the rail stays and the beam is swung another way, or the work waits until an anchor is installed — waiting is one way of satisfying the requirement, not an exception to it. How briefly the opening is exposed makes no difference, workers or signallers below are not fall protection, and logging the removal records a hazard without controlling it.

Q36. Answer: D

ASME B30.9-2021, para. 9-2.10.1(d): when the choker hitch rating is not identified on the sling, the choker hitch rating for single-leg and bridle wire rope slings is 75% of the sling's straight-line hitch rating (70% for cable-laid slings). So 10,000 kg × 0.75 = 7,500 kg — a 25% reduction, caused by the bending and pinching of the rope at the choke point. That 75% figure holds only at an angle of choke of 120° or greater: para. 9-2.10.1(e) sends you to Table 9-2.10.1-1 for smaller angles, which gives 87% of the choker rating at 90°–120°, 74% at 60°–89°, 62% at 30°–59°, and 49% at 0°–29°. Although B30.9 is an American standard, it governs Canadian work because occupational health and safety regulation adopts it by reference — Alberta OHS Code Part 21, s.297(1) requires wire rope, alloy steel chain, synthetic fibre rope, metal mesh and synthetic fibre slings manufactured on or after July 1, 2009 to meet ASME B30.9. If the sling tag carries its own choker rating, use the tag.

Q37. Answer: D

The higher-strength grade carries the higher pretension. Grade A490 has a minimum tensile strength of 150 ksi against 120 ksi for Grade A325, and because the specified minimum pretension is set at 70 percent of the bolt's minimum tensile strength times its tensile stress area, A490 comes out about a quarter higher at every diameter. The RCSC Specification for Structural Joints Using High-Strength Bolts, the United States bolting specification, tabulates it: a 3/4 in bolt is 28 kips for A325 against 35 kips for A490, 7/8 in is 39 against 49, and 1 in is 51 against 64, with A490 higher in every row. The two grades therefore never share a pretension at a given size, and A325 is the lower-strength grade, so it cannot be the stronger one however often it appears on a job. Reuse runs the opposite way from the idea that the higher-strength bolt can be tightened twice. A pretensioned installation stretches the threads between the nut and the thread run-out into the inelastic range, and A490 bolts are not consistently ductile enough to survive a second pretensioning, so RCSC bars any reuse of A490 and of galvanized or coated bolts of either grade; only a plain-finish A325 heavy hex bolt may be reused in a pretensioned joint, and only with the engineer of record's approval. Retightening a bolt that loosened while its neighbours were installed is touching up, not reuse. On Canadian work the design, installation and inspection of bolted joints are governed by CSA S16 for buildings and by CSA S6 or the provincial specification for road bridges; those standards adopt many RCSC provisions but not all of them, and a project specification can be stricter still - Alberta Transportation's Standard Specifications for Bridge Construction has every bolt tensioned once only and permits no reuse at all.

Q38. Answer: B

WorkSafeBC's rigging Part puts it in one line: 'When a sling is applied to a sharp edge of a load, the edge or the sling must be protected to prevent damage to the sling' (Occupational Health and Safety Regulation, Part 15, s. 15.39). Ontario reaches the same hazard from the other side - O. Reg. 213/91 s. 172 (3) provides that no sling or similar device for rigging or hoisting made of web-type fabric or nylon shall be used if it may be cut, and s. 172 (1) (a) requires rigging to be suitable for its intended use. So the plate does not fly on that sling while the edge can cut it. Corner protectors, edge guards or softeners between the edge and the webbing are what removes the exposure.

None of the other three answers takes the edge away. Doubling the sling lays a second ply against the same edge in the same place, and the edge starts on it as soon as the load comes on. Halving the load does not help either, because a cut is not a strength problem: webbing is severed by pressure concentrated over a small radius, which happens at loads far below the rating, and once the fibres are through, the remaining section lets go without warning. Taking the load up slowly still brings the full weight of the plate onto the edge, and the sling then works against it every time the plate swings or settles.

Note what s. 15.39 is written about - a sling, not webbing alone. Chain and wire rope resist a sharp edge far better than polyester does, but the duty to protect the edge or the sling does not change with the material, so choosing a different sling is a decision about what to rig with, not a way of skipping protection at the corner.

Q39. Answer: B

Tight mill scale and light surface rust are acceptable and need not be removed; light rust may slightly improve bond by roughening the surface. Only loose scale and loose, flaking rust have to come off, and the bar must still meet its specified dimensions, mass and deformation height after hand wire brushing. Mill scale and rust are not the same thing: mill scale is the tight blue-black oxide left by hot rolling, while rust is the orange-brown corrosion product that forms afterwards. Canadian project specifications written to CSA A23.1 call for removal of loose scale and loose rust only.

Q40. Answer: A

EL. (Elevation) followed by a number — for example EL. 12500 — gives the finished height of that point in millimetres above the project datum, typically the finished first floor established at EL. 0. So EL. 12500 is 12,500 mm, or 12.5 m, above datum. Elevations above the datum are positive; elevations below it are negative. This is different from a grid coordinate (which locates a point in plan, not in height), a dimension (which measures the distance between two specific points), or a TOS/BOS notation (which gives the elevation of the top or bottom of a specific steel member). When setting anchor bolts or erecting columns, always verify the elevation against the structural drawings — not the architectural drawings, which may use a different datum.

Q41. Answer: D

A PFAS has three essential components: (1) full-body harness, (2) connecting means (lanyard, self-retracting lifeline, rope grab + lifeline), and (3) an anchor point with sufficient strength.

Q42. Answer: D

A drift pin is only a tapered alignment tool for bolt holes. A spud wrench combines a wrench (typically open-end or box) on one end with a tapered shank for alignment on the other, allowing both alignment and bolt tightening in one tool.

Q43. Answer: A

ASME B30.9, para. 9-2.9.5(b)(1): a strand-laid or single-part wire rope sling must be removed from service at 10 randomly distributed broken wires in one rope lay, or 5 broken wires in one strand in one rope lay. Eleven randomly distributed breaks exceeds that limit, so the sling comes out of service immediately — tag it, segregate it so it cannot be picked up again, and report to the rigger-in-charge. Note what the standard does not say: B30.9 sets no broken-wire limit keyed to rope construction, so there is no separate 6×19 or 6×36 figure. Classification governs the minimum clear length of rope between splices (para. 9-2.3.2), not the wire count. The one construction that changes the count is the cable-laid sling, at 20 broken wires per lay. B30.9 is not merely an American reference here — it is adopted by regulation: Alberta OHS Code Part 21, s.297(1) requires wire rope slings manufactured on or after July 1, 2009 to meet ASME B30.9, and s.306 adds further removal triggers, including wear or corrosion affecting individual wires over more than one-third of the original rope diameter, bird-caging, kinking, and heat or arc damage. Beware of carrying figures across: provincial limits for running rope are different and tighter (Alberta s.306(2): 6 randomly distributed broken wires in one rope lay, or 3 in one strand), but a sling is not a running rope. Check the rigging part of your own province's regulation.

Q44. Answer: D

A section that will not close is a drawing conflict, and it goes to the superintendent and the engineer - the ironworker does not settle it at the bar. Run the numbers: 75mm bottom cover, a bottom mat two bars deep (20M is 19.5mm, so 39mm), a top mat another 39mm, and 75mm top cover comes to 228mm, before any chair height, inside a footing shown as 200mm thick. The section as drawn cannot hold what the schedule calls for, so something is wrong - the thickness, the cover, the bar size or the mat callout - and only the designer can say which. Reducing cover, deleting the top mat, or letting the two mats touch each other all change how the footing carries load and how long it lasts in the ground, and none of them is the ironworker's call to make.

Q45. Answer: D

4-M20 A325 means 4 bolts of M20 size (20 mm nominal diameter) meeting high-strength structural bolt requirements. Since 2016 the governing product specification is ASTM F3125 — ASTM A325 and A325M were withdrawn and consolidated into F3125 as grades, so the metric bolt shown here is properly ASTM F3125 Grade A325M, the metric grade covering sizes M12 to M36. Head markings and drawing callouts were deliberately left unchanged, which is why the drawing and the bolt head still read A325 — an apprentice who goes looking for the standalone ASTM A325 standard will find a withdrawn document. On Canadian work the connection itself is governed by CSA S16, Design and construction of steel structures, which sets the resistance of the bolts and the connection and also the rules for installing and inspecting high-strength bolted joints, including pretensioned assemblies. Work to the clause numbering in the edition of that standard referenced by your provincial building code.

Q46. Answer: B

CSA W47.1 (Certification of Companies for Fusion Welding of Steel) governs the certification of welding companies — it certifies the company, not individual welders. Certified companies must have qualified welding supervisors, approved welding procedures (WPSs), and a documented quality control system. This is different from: CSA W59 (which governs the design and fabrication requirements for welded steel structures), CSA W178.2 (which certifies individual welding inspectors), and AWS D1.1 (the American Structural Welding Code used as a reference). On a Canadian structural steel project, the fabricator or erector must hold a current CSA W47.1 certification — this is a mandatory contract requirement.

Q47. Answer: D

100% tie-off means continuous connection — the ironworker must have at least one PFAS connection at all times, including while moving or transitioning their tie-off point. A double-lanyard or SRL with a second connection allows transitioning without a gap.

Q48. Answer: B

Pre-use inspection of rigging hardware is mandatory. Visually inspect for cracks, deformation (hooks opening, shackle bowing), corrosion reducing cross-section, missing or illegible load ratings, and pin security — and verify the WLL is adequate for the lift.

Q49. Answer: A

Camber is a bow along the length of the beam, so it can be read off the piece itself: lay it on level blocking, sight down the top flange from one end, and the curve shows which way the piece rises at midspan. Twenty-five millimetres over a long span is easy to see once you sight it and nearly invisible if you only look at the beam side on. Check the reading against the shop detail for that piece mark before it leaves the ground, then re-mark it. The wrong answers each fail for a different reason. Depth at the ends says nothing about camber, because camber is put into a beam of constant depth by cold cambering or by heat, so both ends measure the same. The mill mark records the producer, the heat number and the grade; it carries no orientation meaning. And 25 mm is not a tolerance to be shrugged off: it is a deliberate quantity the designer put in to be cancelled by deflection under load, so a beam flown upside down starts 25 mm low and finishes about 50 mm out at midspan, which shows up in the floor and in the deck bearing.

Q50. Answer: B

IHSA's Construction Health and Safety Manual chapter on structural steel erection (Ironwork, M036, 2023 edition) gives two controls for column standing: keep the load line vertical and taut while the column is being stood, and keep hands and feet clear of the base plate while the column is being lowered onto the anchor rods. The base plate is the pinch point. A column landing on four rods can shift sideways in an instant, and a hand steadying the plate or a boot bracing it against the footing is between a plate of several hundred kilograms and hard concrete. Control the piece from a distance with taglines instead. Slackening the load line before the column is stable is the other error in this list: the same chapter directs that the column be stable before the rigging is disconnected, and a slack line lets the piece take charge of itself while the crew is still reaching into the connection.

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